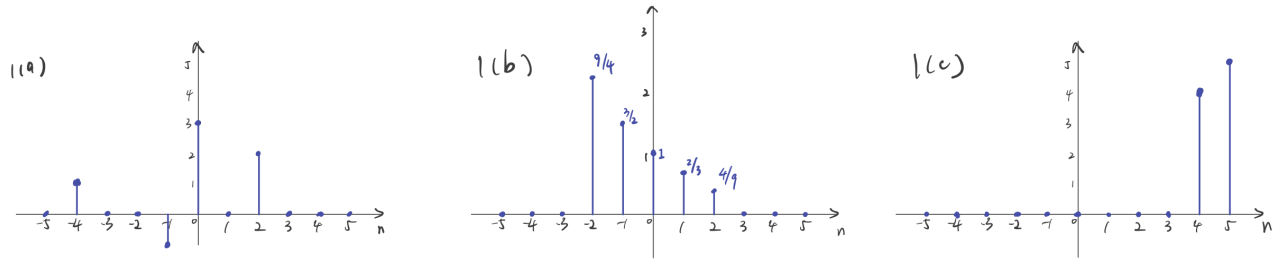
HW Solutions

ECE 310: Digital Signal Processing, Fall 2026 Version: **1.0**

Due Date: 9/4/26
Prepared by: Ethan & Jung Ki

1

Solution:



Grading: *18 points, 6 points each*

- (a), (b), and (c):
  - +6: correct
  - +5: minor mistake in plot (axis labeling, small numerical error, etc)
  - +2: major mistake in plot, effort shown in student's work
  - +0: empty or invalid

2

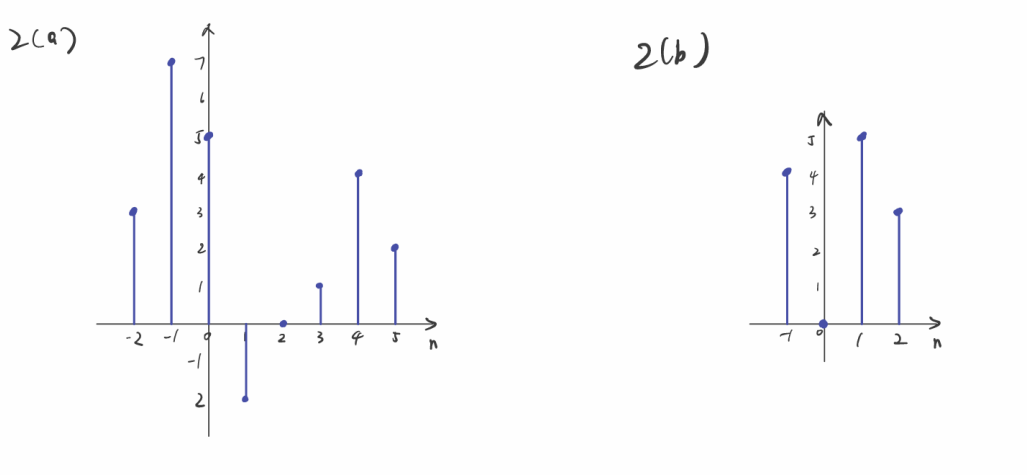
Solution:

a)

$$y[n] = \{3, 7, \underset{\uparrow}{5}, -2, 0, 1, 4, 2\}$$

b)

$$y[n] = \{4, \underset{\uparrow}{0}, 5, 3\}$$



Grading: *14 points, 7 for each (a) and (b)*

- (a) and (b):
  - +7: correct
  - +6: minor mistake in plot (axis labeling, small numerical error, etc)
  - +3: major mistake in plot, effort shown in student's work
  - +0: empty or invalid

3

Solution:

a)

- delta expression: $x[n] = 3(\delta[n+3] + \delta[n+2] + \delta[n+1] + \delta[n] + \delta[n-1])$
- step function expression: $3(u[n+3] - u[n-2])$

b)

- delta expression: $x[n] = 2(\delta[n+3]+\delta[n+2]+\delta[n+1]+\delta[n]) - (\delta[n-1]+\delta[n-2]+\delta[n-3]+\delta[n-4])$
- step function expression: $2u[n+3] - 3u[n-1] + u[n-5]$

Grading: *16 points, 4 for each (i) and (ii)*

- (i):
  - +4: correct Kronecker delta expression
  - +3: minor mistake in Kronecker delta expression
  - +2: major mistake in Kronecker delta expression
  - +0: empty or invalid Kronecker delta expression
- (ii):
  - +4: correct unit-step expression
  - +3: minor mistake in unit-step expression
  - +2: major mistake in unit-step expression
  - +0: empty or invalid in unit-step expression

4

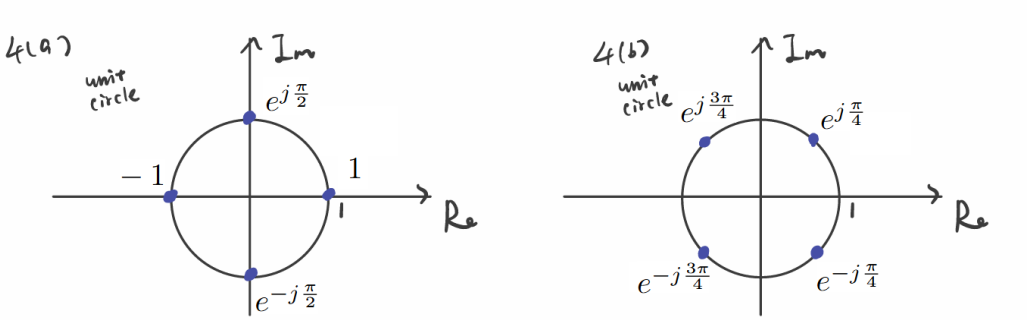
Solution:

a)

$$
\begin{aligned}
z^4 - 1 &= 0 \\
z^4 &= e^{j(2\pi)k}, \quad k \in \mathbb{Z} \\
z &= e^{j\frac{1}{2}\pi \cdot k}
\end{aligned}
$$

b)

$$
\begin{aligned}
z^4 + 1 &= 0 \\
z^4 &= e^{j(-\pi + 2\pi k)}, \quad k \in \mathbb{Z} \\
z &= e^{j\left(-\frac{\pi}{4} + \frac{1}{2}\pi k\right)}
\end{aligned}
$$



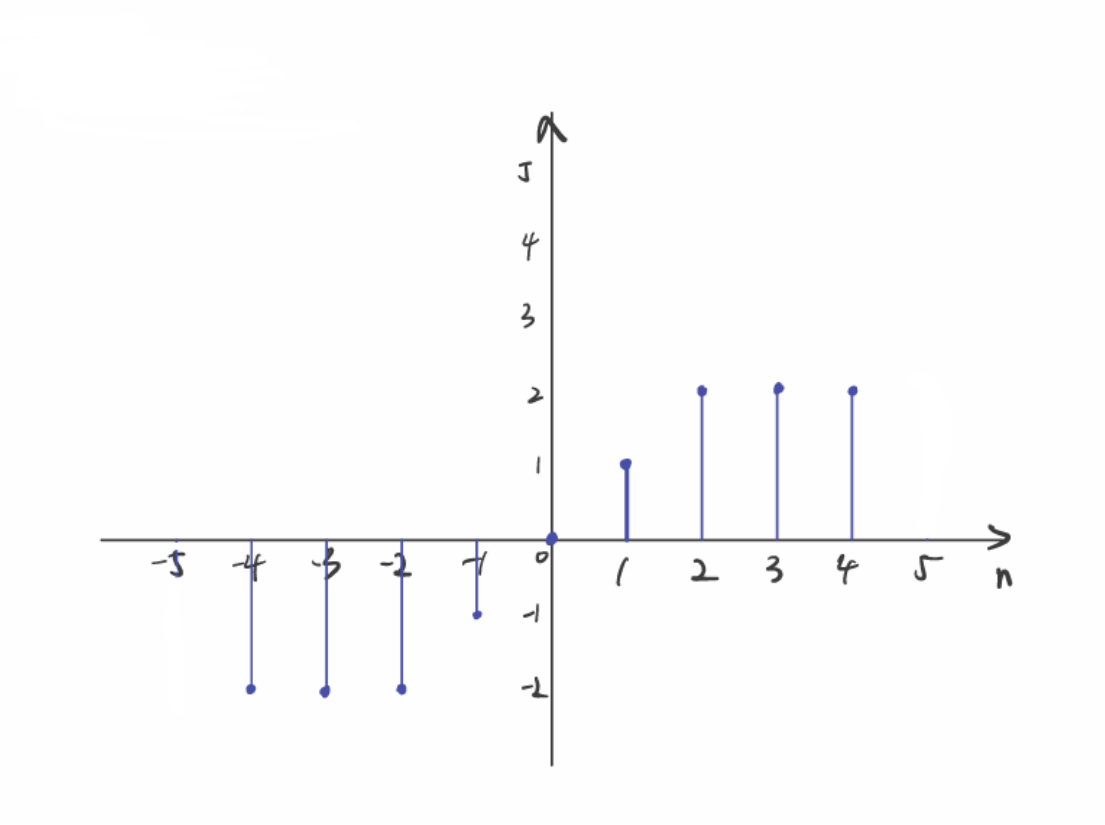
Grading: *16 points, 4 for each (i) and (ii)*

- (i):
  - +4: correct z values
  - +3: minor mistake in z values
  - +2: major mistake in z values
  - +0: empty or invalid in z values
- (ii):
  - +4: correct z values plot
  - +3: minor mistake in z values plot
  - +2: major mistake in z values plot
  - +0: empty or invalid z values plot

5

Solution:

a)



b) It's **NOT LINEAR**. We can take a counter example by letting $x_1[n] = 1, x_2[n] = n^2, a = 0, b = 4$, then we can find

$$T(x_1[n] + x_2[n]) = \{\cdots, 4, 4, 2, \underset{\uparrow}{1}, 2, 4, 4, \cdots\}$$
$$T(x_1[n]) + T(x_2[n]) = \{\cdots, 5, 4, 2, \underset{\uparrow}{1}, 2, 4, 5, \cdots\}$$
$$T(x_1[n] + x_2[n]) \neq T(x_1[n]) + T(x_2[n])$$

c) It's **SHIFT INVARIANT**:

$$T(x[n - n_0]) = \begin{cases} b, & x \geq b \\ x[n - n_0], & a < x[n - n_0] < b \\ a, & x[n - n_0] \leq a \end{cases}$$

$$T(x)[n - n_0] = \begin{cases} b, & x \geq b \\ x[n - n_0], & a < x[n - n_0] < b \\ a, & x[n - n_0] \leq a \end{cases}$$

We can find that the clip operation will shift together with the input signal, so the system is shift-invariant.

Grading: *18 points, 4 for (a), 7 for each (b) and (c)*

- for (a):

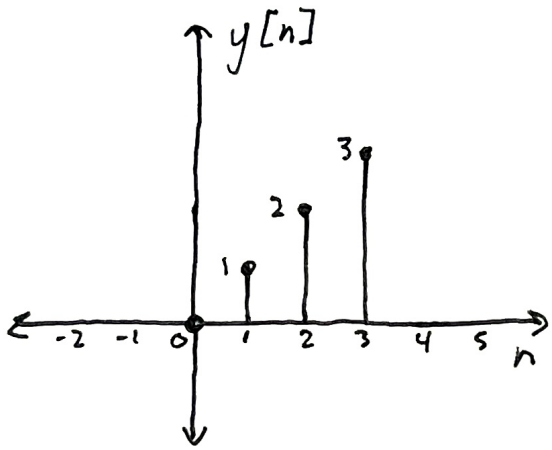
- +4: correct plot
- +3: minor mistake in plot
- +2: major mistake in plot
- +0: empty or invalid

- for (b) and (c):
  - +7: correct choice with correct reasoning
  - +6: correct choice with faulty reasoning
  - +3: incorrect choice with effort clearly shown
  - +0: empty or invalid

6

Solution:

a)



b) It is **LINEAR**:

$$T(ax_1[n] + bx_2[n]) = \begin{cases} ax_1[n] + bx_2[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} aT(x_1[n]) + bT(x_2[n]) &= a\left(\begin{cases} x_1[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}\right) + b\left(\begin{cases} x_2[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}\right) \\ &= \begin{cases} ax_1[n] + bx_2[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases} \\ &= T\left(ax_1[n] + bx_2[n]\right) \end{aligned}$$

c) It is **TIME-VARYING**. Let's take the signal from part (a) as an example, keeping $N = 4$:

Passing $x[n] = \{-2, -1, \underset{\uparrow}{0}, 1, 2, 3, 4, 5, 6\}$ results in the output $y[n] = \{\underset{\uparrow}{0}, 1, 2, 3\}$. If we shift $y[n]$ by $n_0 = 1$, we get

$$y[n-1] = T(x)[n-1] = \{\underset{\uparrow}{0}, 0, 1, 2, 3\}.$$

On the other hand, if we first shift $x[n]$ by $n_0 = 1$ and then pass it through the system, we get

$$T(x[n-1]) = \begin{cases} \{-2, \underset{\uparrow}{-1}, 0, 1, 2, 3, 4, 5, 6\}, & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases} = \{\underset{\uparrow}{-1}, 0, 1, 2\}$$

$$\{\underset{\uparrow}{0}, 1, 2, 3\} \neq \{\underset{\uparrow}{-1}, 0, 1, 2\}$$

Thus, the system is time-varying.

Grading: *18 points, 4 for (a), 7 for each (b) and (c)*

- for (a):
  - +4: correct plot
  - +3: minor mistake in plot
  - +2: major mistake in plot
  - +0: empty or invalid
- for (b) and (c):
  - +7: correct choice with correct reasoning
  - +6: correct choice with faulty reasoning
  - +3: incorrect choice with effort clearly shown
  - +0: empty or invalid